Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

Event Management System

The domain of the Project:
SALESFORCE

Team Mentors (and their designation):

Sriram Yarrabothula

Salesforce Developer

Team Members:

Mr. Dibyajyoti Nayak
Mr. Anurag Singh
Mr. Pappu Kumar JHA
Ms. Ravneet Kaur
Mr. Febin Thomas

Period of the project

November 2025 to April 2026

Declaration

The project titled **"Event Management System"** has been mentored by **Sriram Yarrabothula Sir**, organised by SURE Trust, from **November 2025 to April 2026**, for the benefit of the educated unemployed rural youth for gaining hands-on experience in working on industry relevant projects that would take them closer to the prospective employer. I declare that to the best of my knowledge the members of the team mentioned below, have worked on it successfully and enhanced their practical knowledge in the domain.

Team Members:

Ms. ---------- -- Signature
Mr. Dibyajyoti Nayak -- Dibyajyoti Nayak
Mr. Anurag Singh -- Anurag Singh
Mr. Pappu Kumar JHA -- Pappu Kumar JHA
Ms. Ravneet Kaur -- Ravneet Kaur
Mr. Febin Thomas -- Febin Thomas

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Table of contents

1. Executive summary
2. Introduction
3. Project Objectives
4. Methodology & Results
5. Social / Industry relevance of the project
6. Learning & Reflection
7. Future Scope & Conclusion

Executive Summary

The **"Event Management System"** project was developed to modernize and streamline the end-to-end lifecycle of event planning, execution, and monitoring within organizations using the Salesforce platform.

- Traditional event management approaches often rely on fragmented tools and manual coordination, resulting in inefficiencies, data inconsistencies, and limited visibility.

- This project introduces a unified, scalable, and user-centric solution that enhances operational efficiency, transparency, and data-driven decision-making.

- The system is architected using a robust Salesforce data model, incorporating standard and custom objects with well-defined relationships.

- It enables seamless event creation, attendee registration, vendor allocation, and payment tracking through an integrated workflow.

- Advanced automation using Flows and Apex ensures real-time processing, reduces manual intervention, and enforces business rules effectively.

- Built-in validation mechanisms maintain high data integrity by preventing duplicates, enforcing logical constraints, and ensuring compliance.

- Lightning Web Components deliver a dynamic and responsive user interface, enhancing user engagement and usability.

- Comprehensive reporting and dashboards provide actionable insights into event performance, participation trends, and financial metrics.

- The system significantly reduces processing time, minimizes errors, and improves coordination across all stakeholders.

- It empowers users with real-time status tracking while offering administrators greater control, monitoring, and governance capabilities.

- The platform supports strategic decision-making through structured data analytics and performance visualization.

- Future enhancements may include AI-driven event recommendations, predictive analytics, and integration with external systems for extended functionality.

- Overall, this project demonstrates the effective use of modern cloud technologies to transform event management into an intelligent, automated, and scalable business solution.

Introduction

The Event Management System is a Salesforce-based application designed to streamline and automate the planning, organization, and execution of events within an organization. It provides a centralized platform that integrates all key activities such as event creation, attendee management, vendor coordination, and payment tracking, ensuring a more efficient and structured workflow.

Background and Context:

Event management in many organizations is traditionally handled through manual processes or multiple disconnected tools such as spreadsheets, emails, and standalone applications. This fragmented approach often leads to inefficiencies, data redundancy, lack of coordination, and limited real-time visibility into event operations. With the growing need for organized and scalable solutions, organizations require a centralized system that can efficiently manage all aspects of event planning and execution in a streamlined manner.

Problem Statement / Goals:

The primary objective of this project is to design and implement an integrated Event Management System that simplifies and automates the complete event lifecycle. The system aims to enable efficient event creation, attendee registration, vendor management, and payment tracking while minimizing manual effort and errors. It also focuses on improving data accuracy, enhancing user experience, and providing real-time insights to support better decision-making for stakeholders.

Scope and Limitations:

The scope of the project includes managing events, organizers, attendees, vendors, registrations, and payments within a unified Salesforce platform. It covers key functionalities such as automation, validation, reporting, and dashboard visualization. However, the system is limited to the Salesforce ecosystem and does not currently support external integrations such as third-party payment gateways, mobile applications, or advanced AI-based analytics. These aspects can be considered for future enhancements.

Innovation Component:

The project introduces innovation through the use of Salesforce technologies such as Flows for process automation, Apex for advanced business logic, and Lightning Web Components for a dynamic and interactive user interface. It combines a well-structured data model with automation and real-time analytics to deliver a smart, scalable, and user-friendly solution. This approach transforms traditional event management into a more efficient, automated, and data-driven process.

Project Objectives

- **Objectives and Goals**

1. To design and develop a centralized Event Management System that streamlines the complete event lifecycle, including event creation, attendee registration, vendor coordination, and payment tracking, while reducing manual effort and improving operational efficiency.
2. To implement automation and validation mechanisms using Salesforce tools such as Flows and Apex to ensure data accuracy, minimize errors, and enable real-time processing of event-related activities.
3. To provide a secure, user-friendly, and interactive platform for different stakeholders (organizers, attendees, administrators) with controlled access, seamless navigation, and improved user experience.
4. To enable effective decision-making by offering real-time insights through reports and dashboards, allowing better monitoring of event performance, participation, and financial aspects.

- **Expected Outcomes and Deliverables**

1. A fully functional and scalable Event Management System developed on Salesforce, capable of efficiently managing multiple events and related operations in a structured manner.
2. A well-defined data model with standard and custom objects, along with automated workflows, validation rules, and Apex logic to support core functionalities.
3. Interactive dashboards, reports, and user interfaces (LWC) that provide clear visibility into event data, improve transparency, and enhance overall system usability.
4. A significant improvement over traditional manual systems in terms of efficiency, accuracy, data management, and overall coordination, along with proper project documentation for future enhancements.

Methodology and Results

• Methods / Technology Used

The project adopts a systematic and iterative development methodology on the Salesforce platform, combining both declarative and programmatic approaches. Core functionalities are implemented using standard Salesforce features such as objects, relationships, validation rules, and security settings, while advanced requirements are handled using Apex (triggers, classes) and Flows for automation. Lightning Web Components (LWC) are used to build a responsive and interactive user interface, ensuring better usability and performance.

The development process includes requirement analysis, data modeling, implementation, testing, and deployment. Each module (event creation, registration, vendor management, payment tracking) is developed and tested individually before integration into the complete system.

• Tools / Software Used

The project was developed using the following tools and technologies:

1. Salesforce Lightning Platform
2. Lightning Web Components (LWC)
3. Apex Programming Language
4. Salesforce Flow Builder
5. SOQL (Salesforce Object Query Language)
6. Salesforce Reports & Dashboards

These tools enabled the development of a scalable, secure, and efficient system.

• Data Collection Approach

The system collects and manages data entered by users, such as event details, attendee information, vendor assignments, and payment records. Input is captured through forms and validated using rules to ensure accuracy and consistency. The data is stored in structured objects and is used for tracking, reporting, and decision-making purposes.

• Project Architecture

The system follows a multi-layered architecture within the Salesforce ecosystem:

1. **Data Layer:**
   Includes standard objects (Account, Contact) and custom objects (Event, Registration, Vendor, Payment, Event-Vendor). Relationships such as lookup and master-detail ensure proper data linkage and maintain data integrity.
2. **Application Layer:**
   Implements business logic using Flows and Apex. This layer manages automation such as

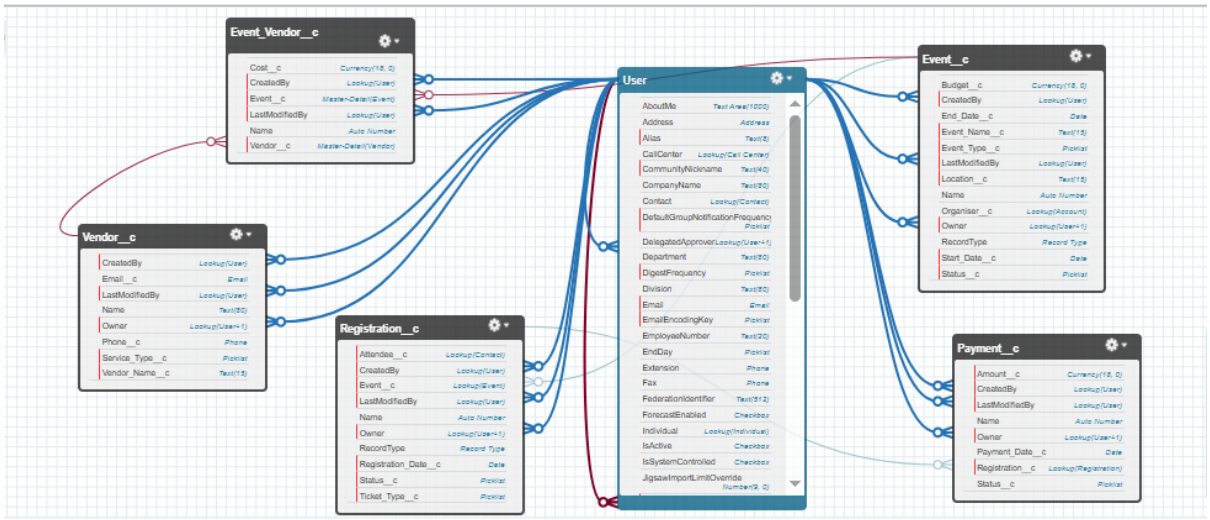
registration processing, payment updates, duplicate prevention, and event status management.

3. **Presentation Layer:**
   Built using Lightning Web Components (LWC) and Lightning App Builder, providing an intuitive and interactive user interface for different users.
4. **Security Layer:**
   Managed using Profiles, Roles, and Sharing Rules to ensure data privacy and controlled access based on user roles.
5. **Analytics Layer:**
   Reports and Dashboards provide real-time insights into event performance, revenue, attendee participation, and vendor costs.

This layered architecture ensures scalability, flexibility, and efficient system performance.

Salesforce Data Model (Objects and Relationships)

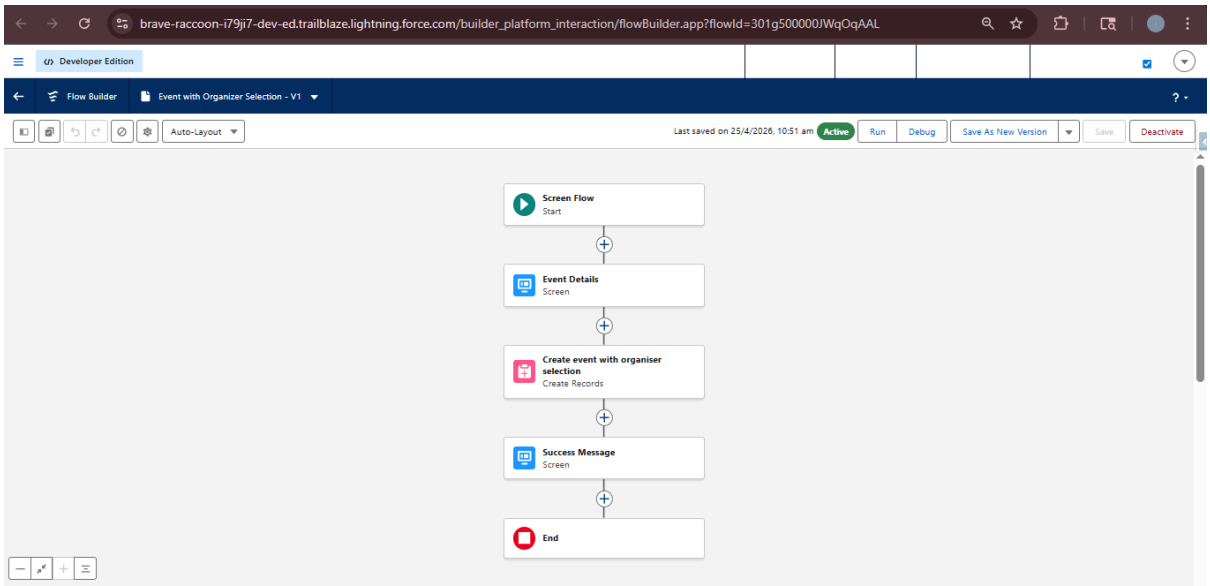
This below figure represents a Salesforce Event management System where **User** acts as the central object connected to all other objects. The **Event__c** object is linked with **Registration__c** to manage attendees, and **Registration__c** is further connected to **Payment__c** to handle payments. The **Vendor__c** object is connected to **Event__c** through the **Event_Vendor__c** junction object, creating a many-to-many relationship between events and vendors. Overall, the connections show a clear flow where events are associated with registrations and payments, while vendors are linked to events through a junction object, making the system well-structured and organized.



Automation using Salesforce Flows

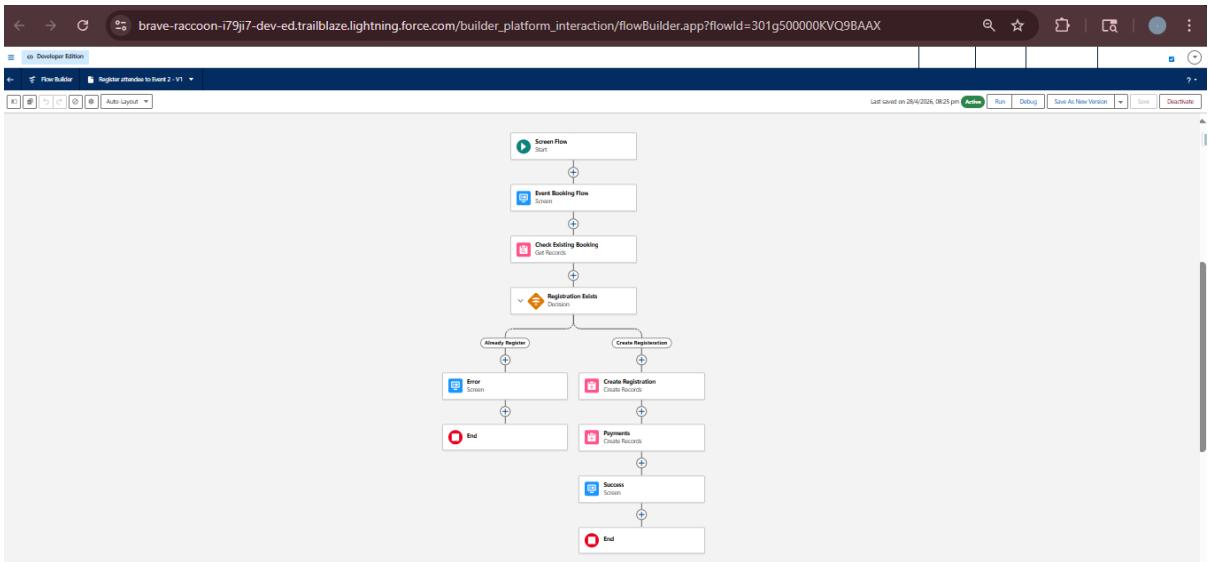
1. Event Creation Flow (Screen Flow)

This flow allows organizers to create new events by entering details such as event name, date, location, and budget through a user-friendly interface.Once submitted, the system automatically stores the data and confirms successful event creation.



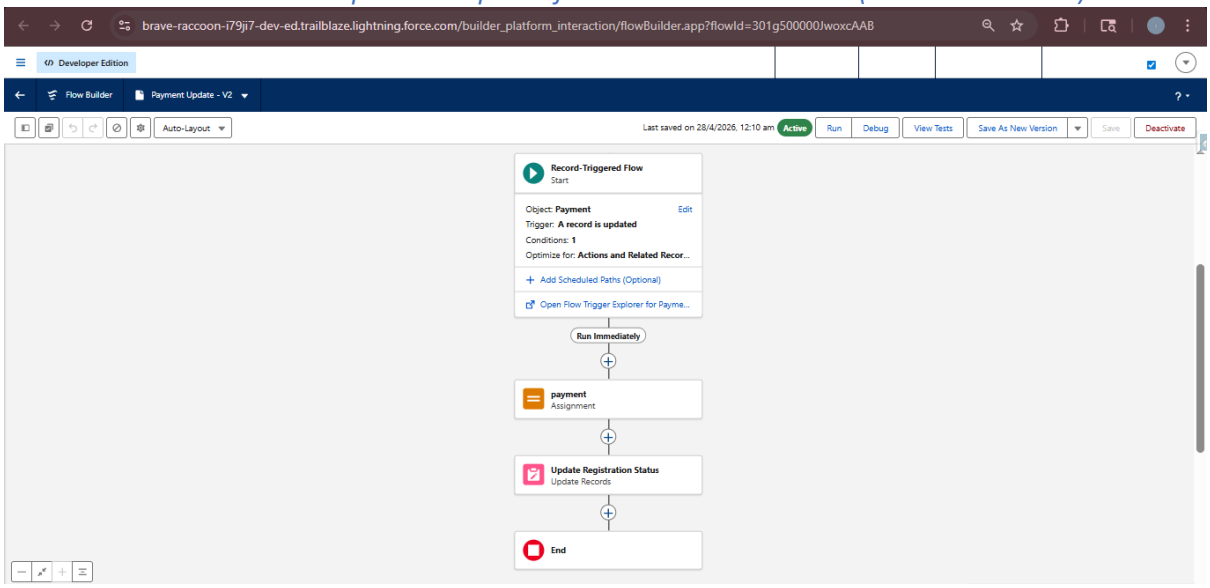
2. Attendee Registration Flow (Screen Flow)

This flow enables users to register for events by selecting an event and entering attendee details.It validates inputs, prevents duplicate registrations, and creates a registration record linked to both the event and the attendee.



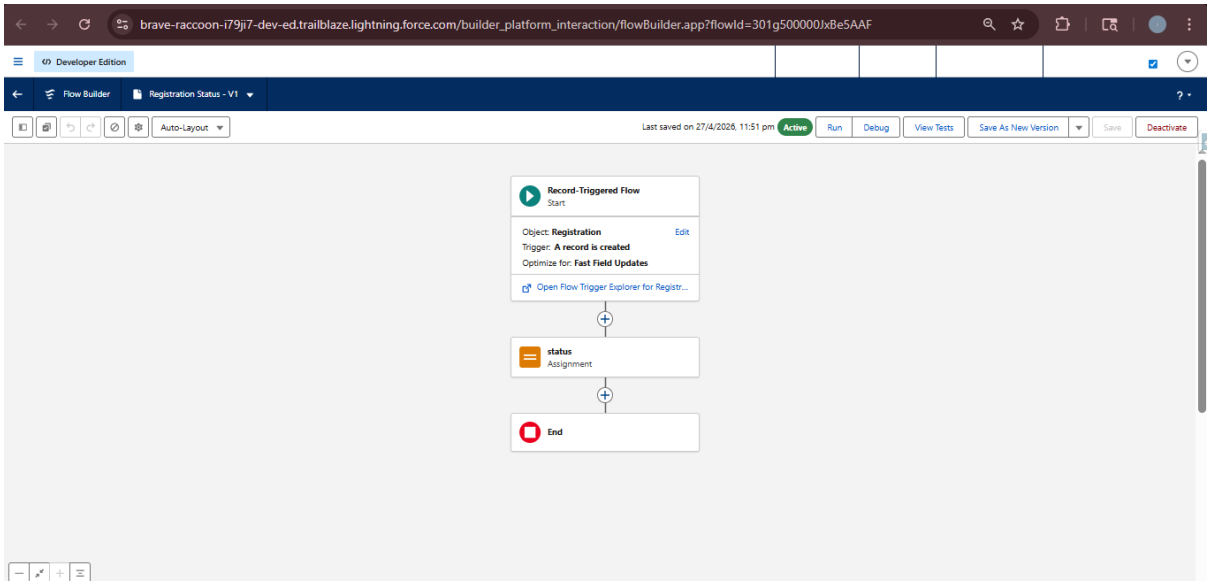
3. Payment to Registration Status Flow (Record-Triggered Flow)

This flow is activated whenever a Payment record is created or updated.It evaluates the payment status and automatically updates the related Registration status to “Confirmed” when payment is completed.



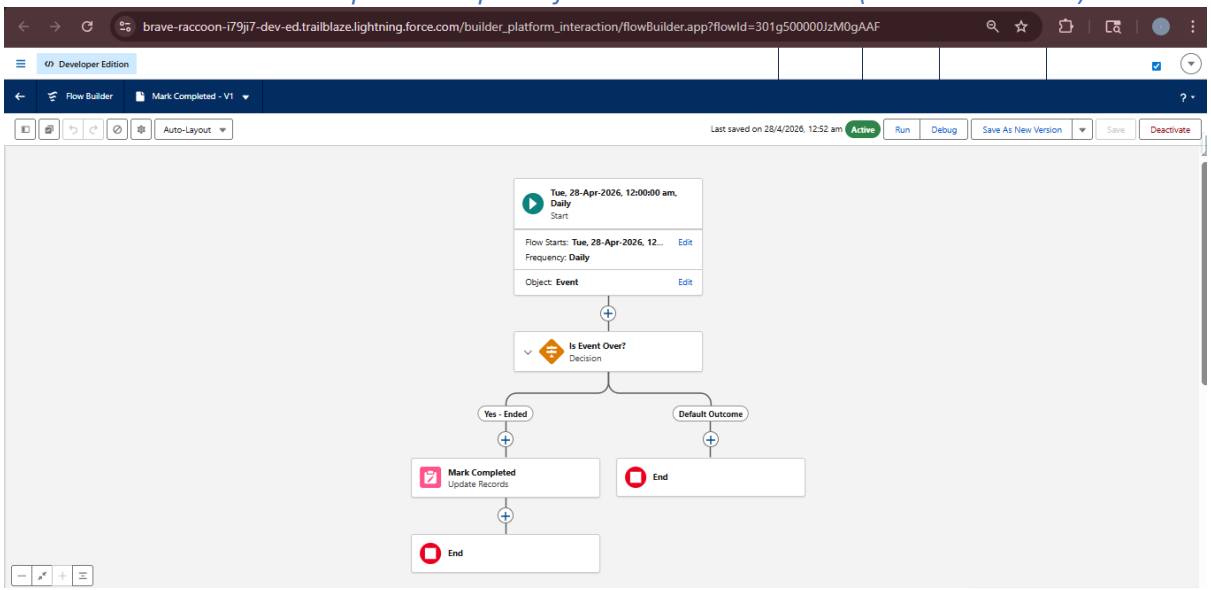
4. Event Status Update Flow (Record-Triggered Flow)

This flow is triggered when an Event record is created or updated.It compares event dates with the current date and updates the event status to Upcoming, Ongoing, or Completed.



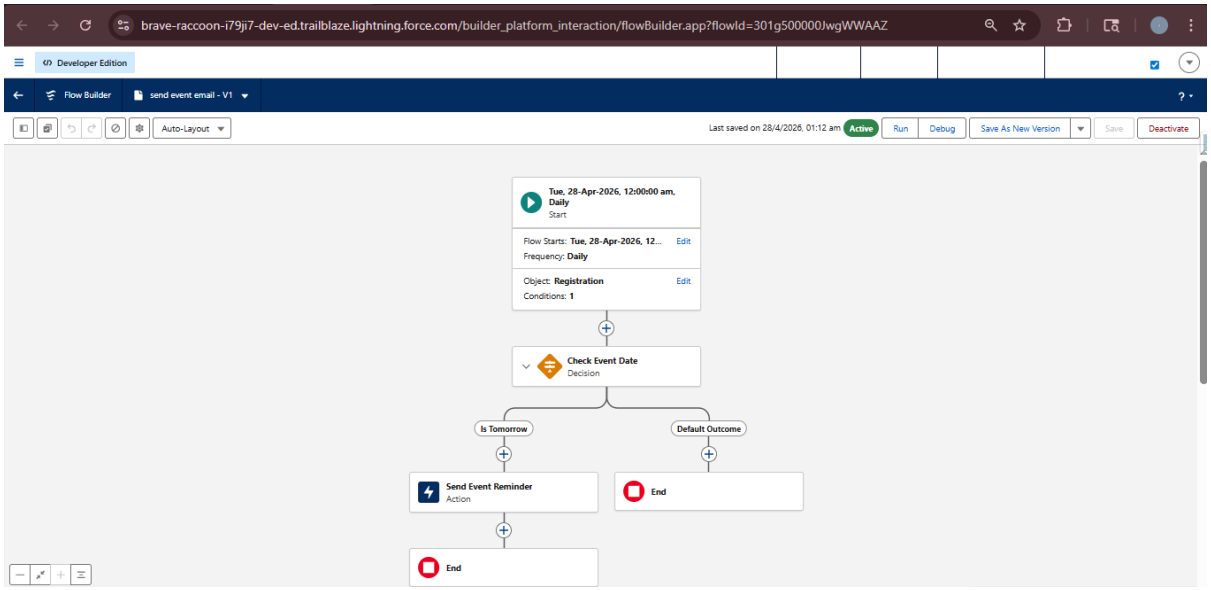
5. Event Status Scheduled Flow

This scheduled flow runs at defined intervals (such as daily).It automatically updates event statuses based on their start and end dates to keep records accurate.



6. Event Reminder Scheduled Flow

This scheduled flow sends automated reminders to attendees and organizers before the event date. It triggers notifications or emails to improve communication and participation.



□ User Roles and Security Management

The system ensures data security and controlled access using Salesforce security features. The **Organization-Wide Default (OWD)** is set to *Private*, which restricts record access to only the owner and authorized users, ensuring data confidentiality.

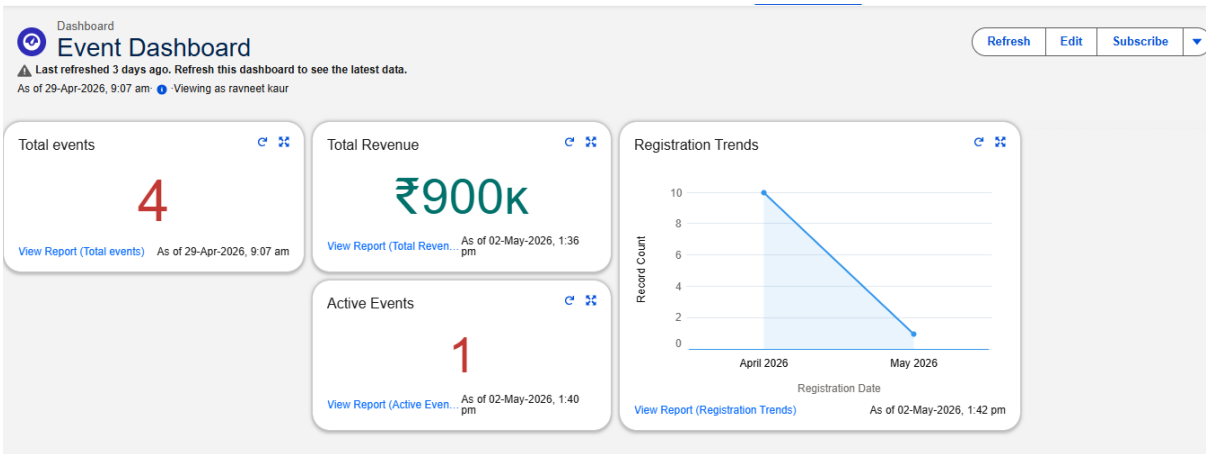
A **Role Hierarchy** is implemented with levels such as *Admin, Event Manager, Organizer, and Attendee*, allowing higher roles to access and manage data of lower roles, ensuring proper supervision and control.

Different **Profiles** like *System Admin, Event Manager, and Vendor User* are defined to provide role-based permissions, controlling what actions users can perform within the system. This combination of OWD, roles, and profiles ensures secure, structured, and efficient access management.

Final project working screenshots along with supporting explanation

1 . <u>Dashborad</u>

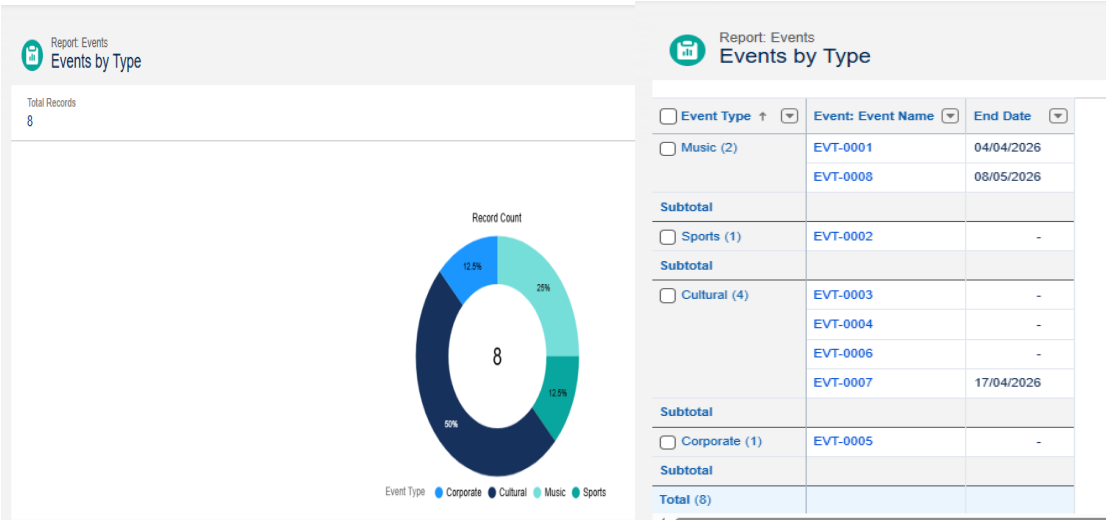
The dashboard provides a visual and real-time overview of key system metrics. It displays important information such as Total Events, Total Revenue, Active Events, and Registration Trends, allowing users to quickly understand overall performance and track progress efficiently.



2.**<u>Report</u>**

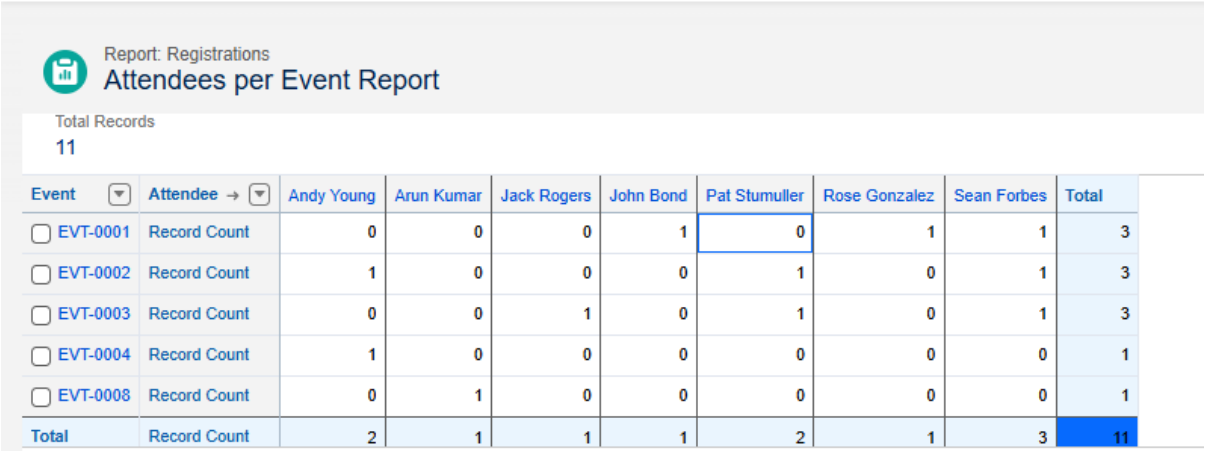
a. **Event By type**

The **Events by Type report** categorizes events based on their type, helping users understand the distribution of different kinds of events and analyze which type is most frequent.



b. Attendees Per Event

The **Attendees per Event report** provides a detailed view of how many attendees are registered for each event. This helps in tracking participation levels, evaluating event popularity, and making improvements in event planning and promotion.



Report: Registrations
Attendees per Event Report

Total Records
11

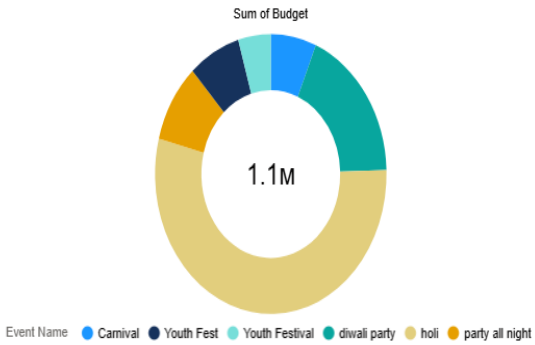
Event	Attendee	Andy Young	Arun Kumar	Jack Rogers	John Bond	Pat Stumuller	Rose Gonzalez	Sean Forbes	Total
EVT-0001	Record Count	0	0	0	1	0	1	1	3
EVT-0002	Record Count	1	0	0	0	1	0	1	3
EVT-0003	Record Count	0	0	1	0	1	0	1	3
EVT-0004	Record Count	1	0	0	0	0	0	0	1
EVT-0008	Record Count	0	1	0	0	0	0	0	1
Total	Record Count	2	1	1	1	2	1	3	11

c. Total Revenue

The **Revenue report** summarizes the total income generated from event registrations and payments. It helps in monitoring financial performance, comparing revenue across events, and supporting better financial planning and decision-making.

Report: Events
Total Revenue

Event Name	Event: Event Name	Budget
Carnival (1)	EVT-0003	₹70,000
Subtotal		₹70,000
diwali party (1)	EVT-0008	₹2,00,000
Subtotal		₹2,00,000
holi (1)	EVT-0004	₹6,00,000
Subtotal		₹6,00,000
party all night (1)	EVT-0005	₹1,00,000
Subtotal		₹1,00,000
Youth Fest (1)	EVT-0002	₹80,000
Subtotal		₹80,000
Youth Festival (1)	EVT-0007	₹50,000
Subtotal		₹50,000
Total (6)		₹11,00,000



d. Vendor Per Cost

The **Vendor Cost report** highlights the expenses incurred for vendors in each event, such as catering or decoration costs. It enables organizers to manage budgets efficiently, control expenses, and analyze cost distribution across different events.

Report: Events with Event Vendors and Vendors
Vendor Cost Report

Total Records
1

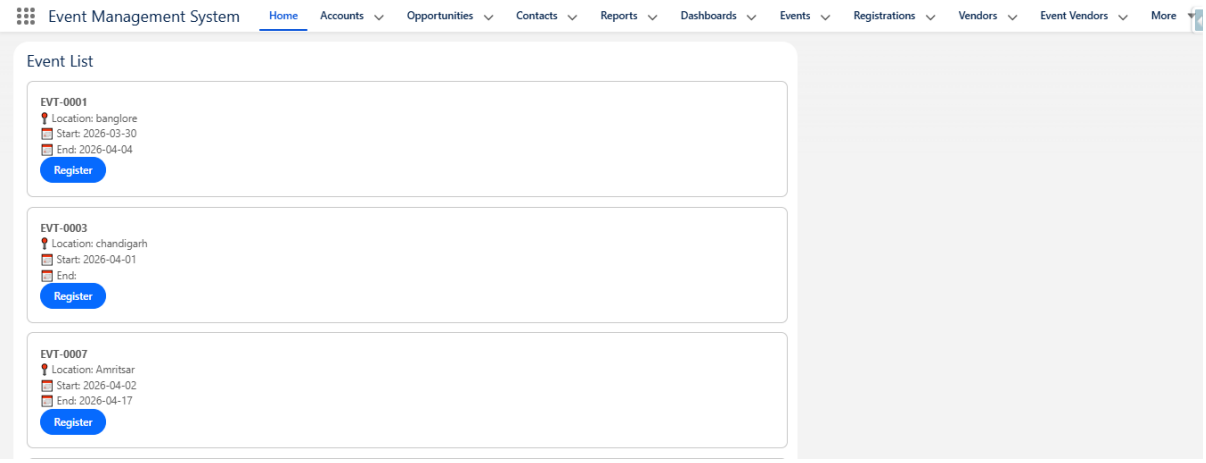
Event Name	Budget	₹2,00,000	Total
diwali party	Record Count	1	1
Total	Record Count	1	1

Details (1 Rows) Click an intersection in the table above to filter details.

	Event: Event Name	Vendor: Vendor Name	Cost
1	EVT-0008	Arun kumar	₹40,000
2			

3. LWC COMPONENTS

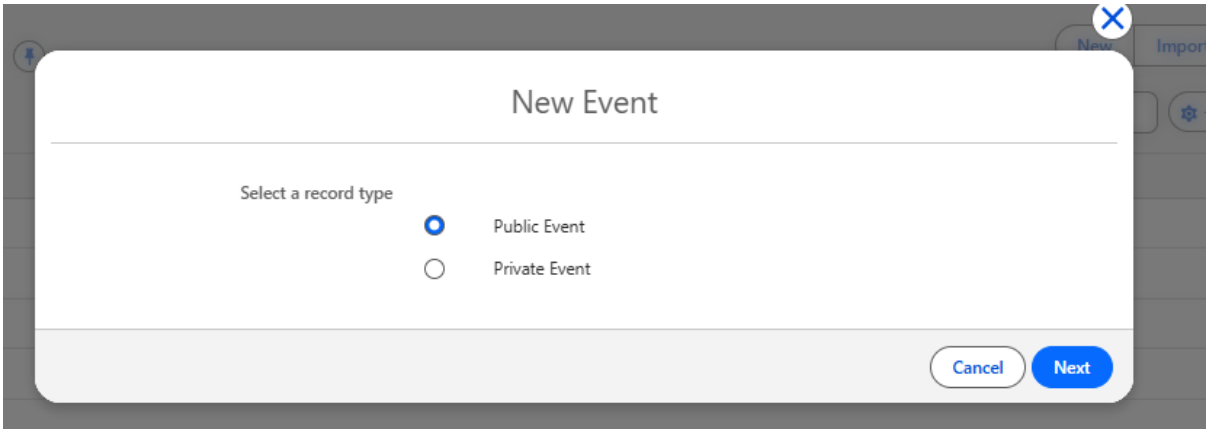
The **Event Dashboard** is a Lightning Web Component (LWC) in Salesforce that provides a clear and centralized view of all events in the system. It displays important details such as event name, location, start date, end date, and status, allowing users to quickly understand what events are upcoming or ongoing. The dashboard retrieves data from Salesforce using Apex and presents it in a simple and organized format using Lightning Design System. It helps users and administrators easily monitor events, improve planning, and get a quick summary of all event activities in one place.



4. RECORD CREATION SCREENS

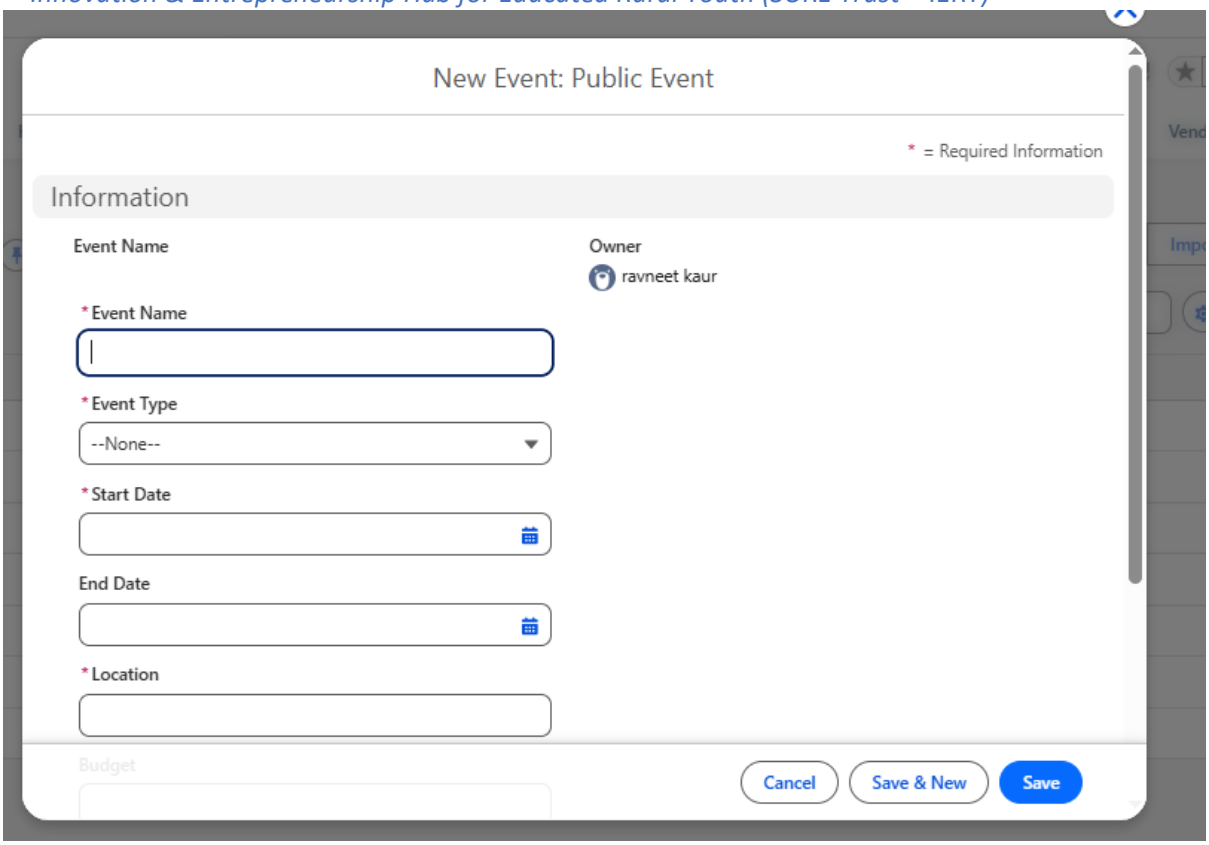
a. Event Type

The event is an **Online and Private event**, which means it is conducted through digital platforms and is accessible only to selected users. This ensures restricted access and better control over participants.



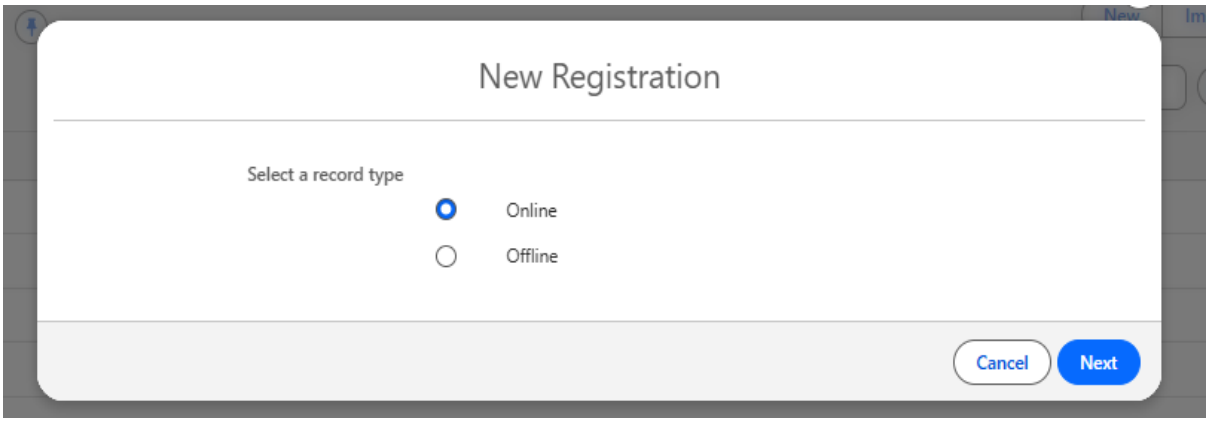
b. Event Type Form

The event form is used to create and manage event details such as event name, location, start date, end date, and record type. It helps in defining whether the event is public or private and stores all necessary event information in the system.



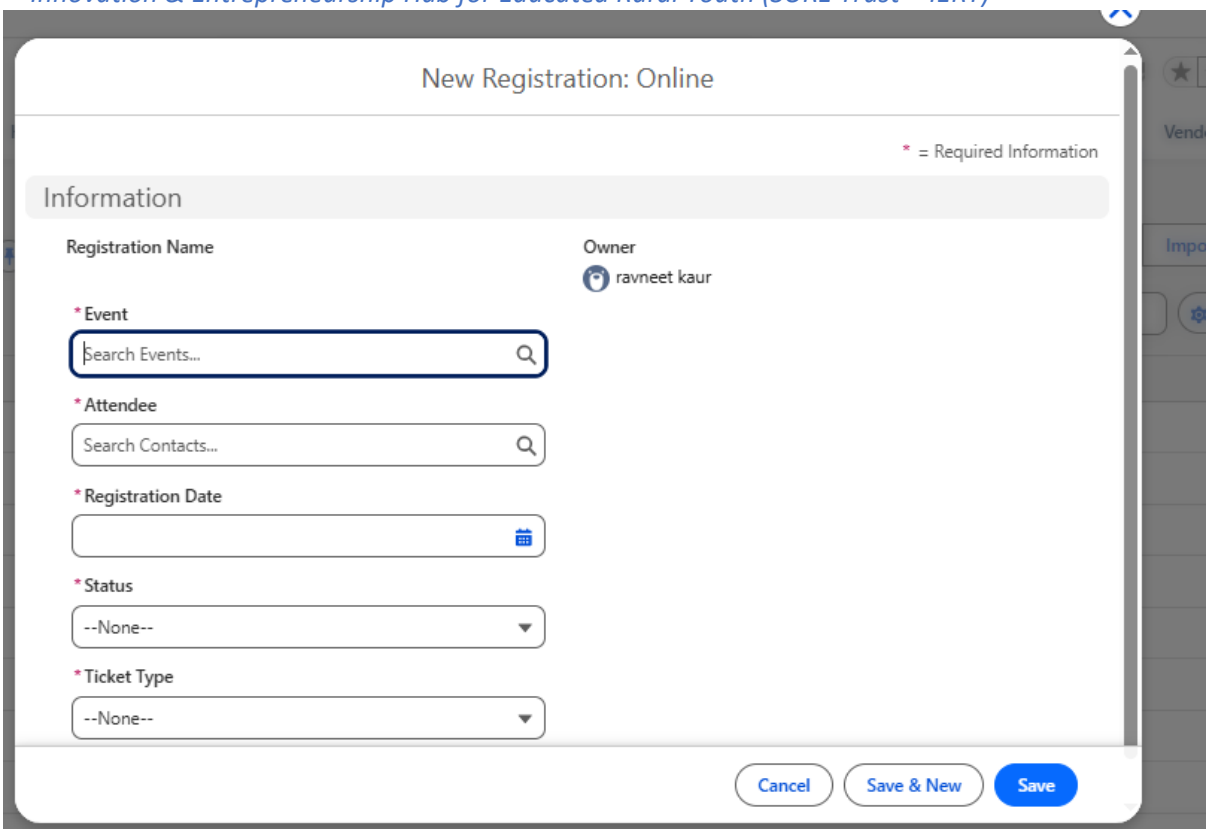
c. Registration Type

Registration can be done in two ways: **Online and Offline**. Online registration is done through the system by users themselves, while offline registration is manually entered by the administrator. This helps in tracking how participants are joining the event.



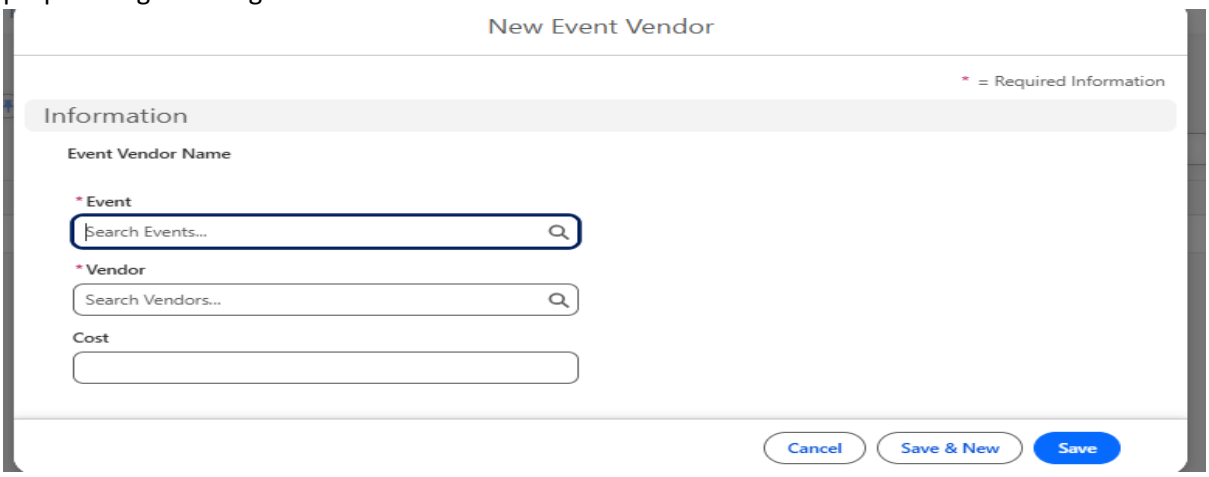
d. Registration Form

The registration form is used to capture participant details such as event selection, contact information, ticket type, and registration mode.



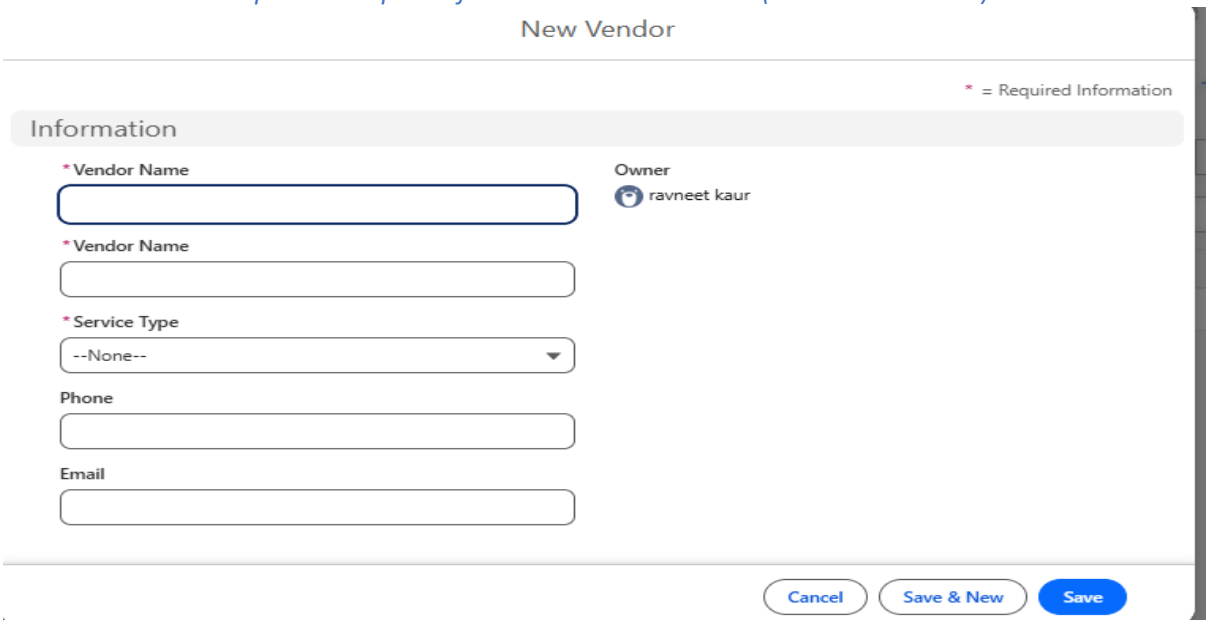
e. Event Vendor Form

The **Event_Vendor__c object** acts as a junction object that connects Events and Vendors, where both Event and Vendor are linked using Master-Detail relationships. It also includes a Cost field to track the expense associated with each vendor assigned to an event, helping in proper budget management.



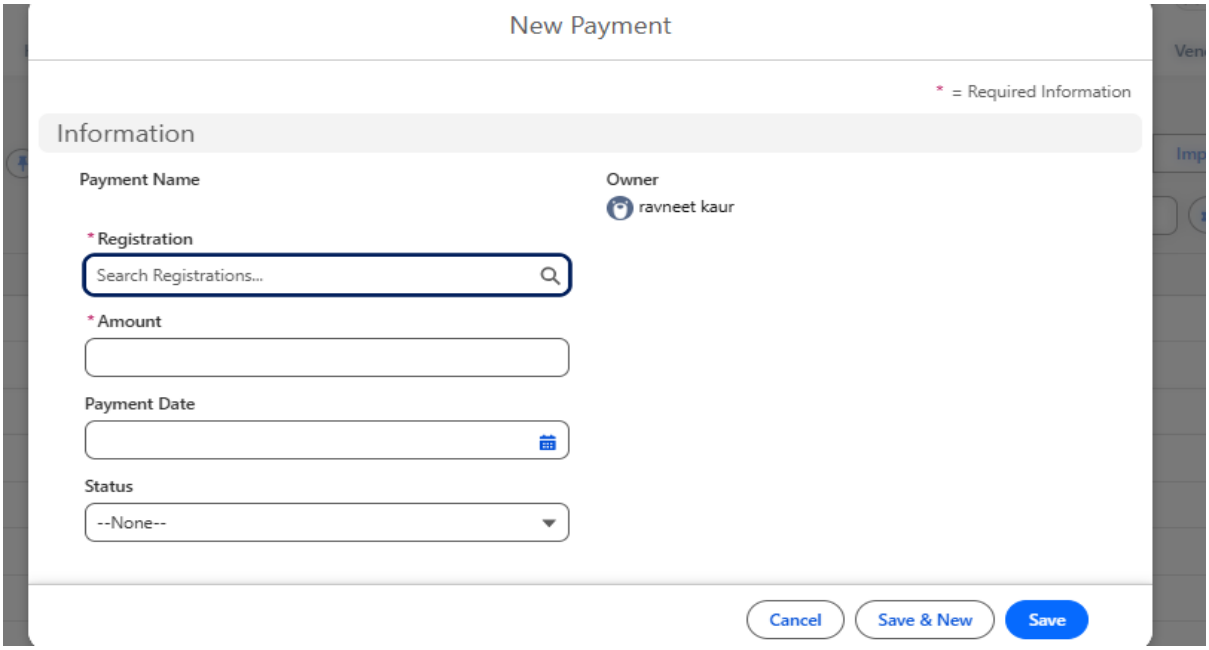
f. Vendor Form

The **Vendor__c object** is used to store details of service providers involved in events, such as catering, lighting, decoration, and sound. It help in maintaining complete vendor contact and service information.



g. Payment Form

The **Payment__c object** is used to manage payments related to event registrations. It contains a lookup to Registration, along with fields like Amount, Payment Date, and Status (Paid or Pending).



Social / Industry relevance of the project

The Event Management System holds strong relevance in both social and industry contexts as it addresses the growing need for digital transformation in event planning and administration. In today's fast-paced environment, organizations, educational institutions, and corporate sectors regularly conduct events such as conferences, workshops, seminars, cultural programs, and training sessions. Managing these activities manually often leads to inefficiencies, miscommunication, and lack of transparency. This project provides a centralized, automated, and structured solution that significantly improves the way events are organized and executed.

From a **social perspective**, the system enhances user convenience by simplifying event registration and participation. Attendees can easily register, select ticket types, and receive updates in real time, which improves engagement and accessibility. It also promotes better communication between organizers and participants through automated notifications and status updates. By reducing paperwork and manual coordination, it contributes to a more sustainable and eco-friendly approach to event management.

From an **industry perspective**, the project aligns with modern enterprise requirements for automation, data accuracy, and real-time analytics. Organizations increasingly rely on platforms like Salesforce to manage business processes efficiently, and this system demonstrates how CRM tools can be extended beyond sales into event management. Features such as Flows, Apex automation, role-based security, and dashboards reflect real-world industry practices used in enterprise applications.

The system also supports **better decision-making and resource optimization** by providing insights into attendance trends, revenue generation, and vendor costs. This is highly valuable for businesses and institutions that aim to improve profitability and operational efficiency. Moreover, it reduces dependency on manual labor, minimizes errors, and ensures faster processing of event-related activities.

Overall, the project demonstrates how digital solutions can transform traditional event management into a more efficient, transparent, and scalable system, making it highly relevant for both social use and industry-level adoption

Learning and Reflection

- **Document every team member's new learnings (in terms of technology, management, etc)**

1. Salesforce Platform Knowledge
Gained hands-on understanding of Salesforce architecture including objects, relationships, Flows, Apex, and Lightning Web Components.

2. Data Modeling Skills
Learned how to design efficient data models using standard and custom objects with proper relationships like lookup and master-detail.

3. Automation Techniques
Developed knowledge of process automation using Screen Flows, Record-Triggered Flows, and Scheduled Flows to reduce manual work.

4. Apex Development
Improved programming skills by writing Apex triggers and classes for implementing advanced business logic and validations.

5. UI Development (LWC)
Learned how to create interactive and responsive user interfaces using Lightning Web Components.

6. Reports and Dashboards
Gained experience in building reports and dashboards for data analysis and business decision-making.

7. Security and Access Control
Understood role hierarchy, profiles, and OWD settings to manage data security and user access.

8. Team Collaboration Skills
Improved communication, task distribution, and coordination while working as a team on a real-time project

- **Document every team member's experience. (overall experience)**

1. Practical Project Experience
Worked on a complete end-to-end Salesforce project, from requirement analysis to deployment, gaining real industry exposure.

2. Problem-Solving Experience
Faced and resolved real-time challenges in flows, relationships, and automation logic, improving analytical thinking.

3. Development Experience
Experienced building a fully functional system with multiple modules like events, registration, vendors, and payments.

4. Team Coordination Experience
Learned how to collaborate effectively with team members, divide tasks, and complete project milestones on time.

5. Technical Confidence

Built confidence in using Salesforce tools and technologies independently for development and customization.

6. Time Management Experience

Improved ability to manage time efficiently while balancing development, testing, and documentation tasks.

7. Professional Exposure

Gained understanding of how real-world business applications are designed and implemented in organizations.

8. Overall Learning Experience

Developed both technical and professional skills, making the project a valuable learning experience in Salesforce development and teamwork.

Conclusion and Future Scope

- **Recap objectives and achievements**

  The primary objective of the Event Management System was to design and develop a centralized platform that efficiently manages the complete event lifecycle, including event creation, attendee registration, vendor coordination, and payment tracking. The system successfully achieved this by implementing a structured Salesforce data model, automation using Flows and Apex, and an interactive user interface using Lightning Web Components. It also ensured data accuracy, reduced manual effort, and improved overall operational efficiency through validation rules and role-based security. Additionally, reports and dashboards were implemented to provide real-time insights into event performance, revenue, and participation, supporting better decision-making.

- **Future scope of this project**

  The project can be further enhanced by integrating mobile application support to allow users to manage events on the go. Advanced features such as AI-based event recommendations, predictive analytics for attendance and revenue forecasting, and chatbot support for user queries can also be added. Integration with external systems like payment gateways, email marketing tools, and calendar applications would further improve functionality. Additionally, incorporating real-time notifications, advanced analytics dashboards, and machine learning-based insights can make the system more intelligent, scalable, and industry-ready.